Axis Communications Family Integration Guide

Axis Communications Family Integration Guide for GCX-ONE (NXGEN Platform)

Introduction

This document describes the end-to-end integration between the NXGEN Monitoring Platform and the Axis Communications family of devices, including:

Axis Network Cameras (IP Cameras)

Axis Network Audio Systems (Horn Speakers)

Axis I/O Relay Modules

Axis Camera Station Pro (ACS)

The purpose of this integration is to enable unified monitoring, event ingestion, device control, audio actions, and simplified configuration of Axis devices from within the NXGEN ecosystem.

This ensures:

Consistent alarm delivery

Automated workflows

Seamless video access

Unified I/O and audio control

Scalable device management

System Overview

Axis devices communicate using a combination of industry-standard and Axis-native protocols:

NXGEN consumes Axis data for:

Live video streaming

Event/alarm notifications

I/O status changes

Audio announcements

ACS stored video playback

System Architecture

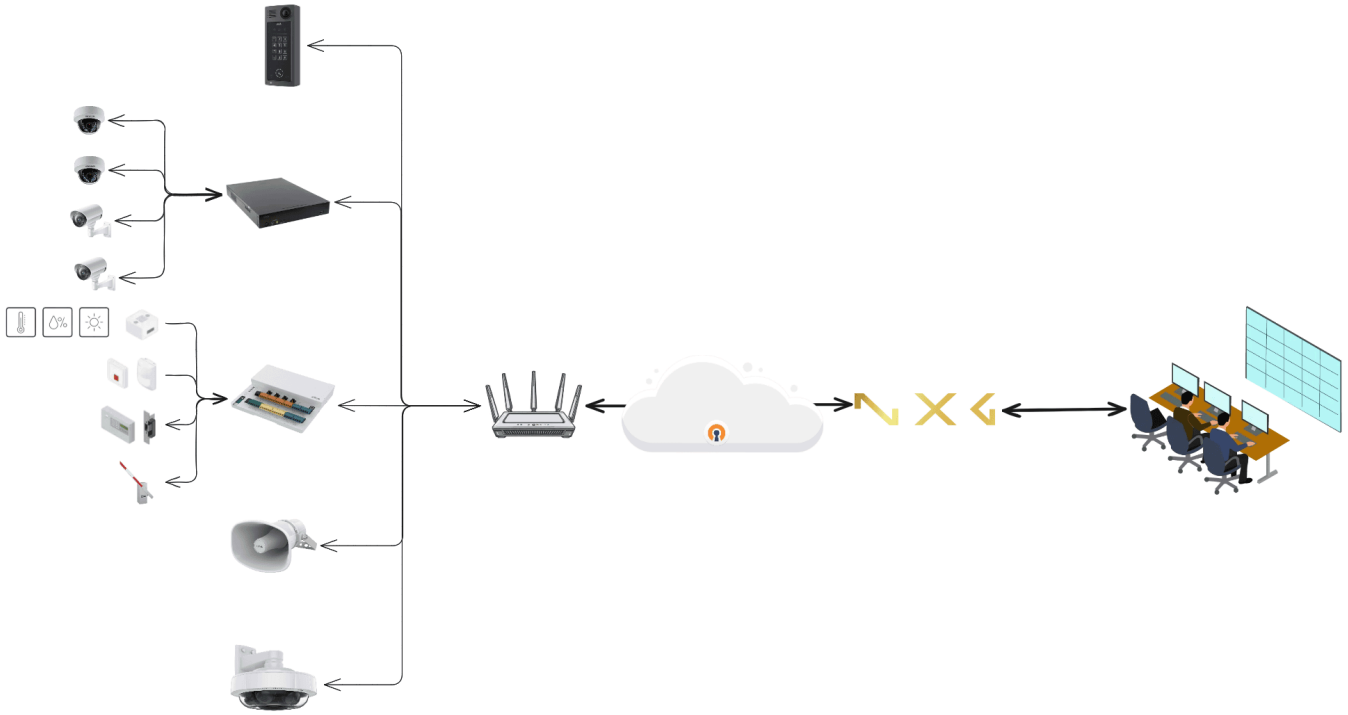
Axis cameras, audio devices, ACS Pro, and I/O Modules connect locally to the site router, which securely links to the cloud via VPN (preferred) or non-VPN. NXGEN receives all video streams, alarms, and device telemetry from the cloud endpoint and presents them to monitoring operators across various applications. The architecture enables:

Centralized management

Unified event handling

Real-time audio and I/O controls

Scalable multi-site monitoring



Communication Flows

Configuration Overview

Pre-Integration Requirements

Axis device updated to latest firmware.

HTTP enabled.

API access user with necessary privileges.

“Web Client” enabled

Ports 29202 (TCP), 29204 (API), and 29205 (streaming) are open (depending on version),80 and 443.

For ACS Pro:

Ensure network connectivity (either public network access or through VPN) between the Site network and the monitoring platform (GCX-ONE).

Enable **VAPIX API** (default enabled).

API Port is open and accessible.

Enable **Event Service**.

Configure **RTSP** parameters if needed (codec, bitrate) to match your network bandwidth.

For IP Cameras:

Device Configuration in the GCX-ONE Platform

Please find the steps below to add Axis Camera Station (ACS) Pro, Axis I/O Modules, and Axis Network Horn Speakers into the GCX-ONE Environment.

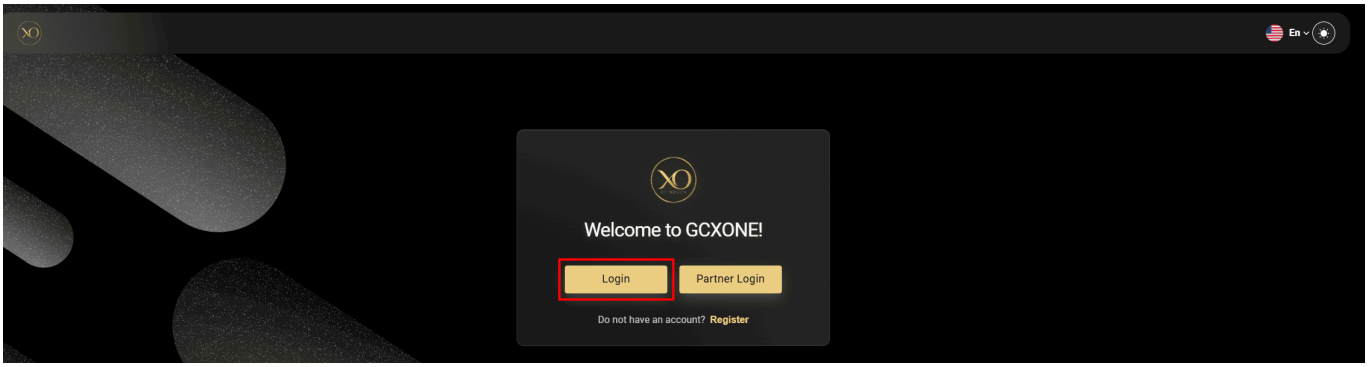
Axis Family devices Configuration

Step 1: Login to GCX-ONE Platform

Open the (GCX-ONE Cloud Experience) URL in a browser.

Sign in using your administrator or integrator credentials.

After successful login, the system initializes your session and loads the environment associated with your tenant.

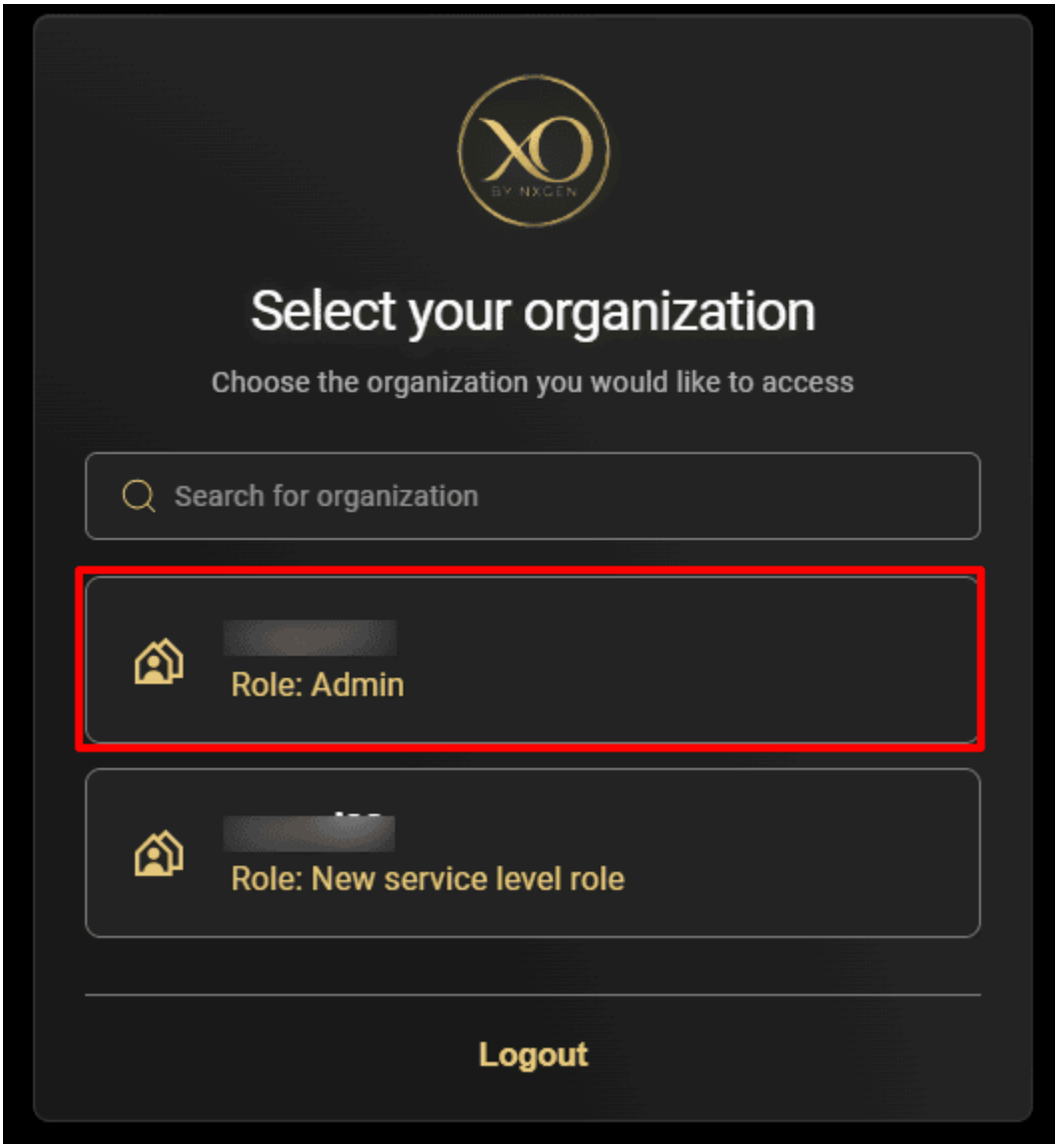


Step 2: Select Your Organization

If your tenant is configured with **multiple organizations**, GCX-ONE will prompt you to select the organization you want to manage.

****If multiple organizations exist: ****You will see an organization selector screen. Choose the correct organization to proceed.

****If only one organization exists: ****GCX-ONE automatically logs you into the **default organization** and directly loads the dashboard.

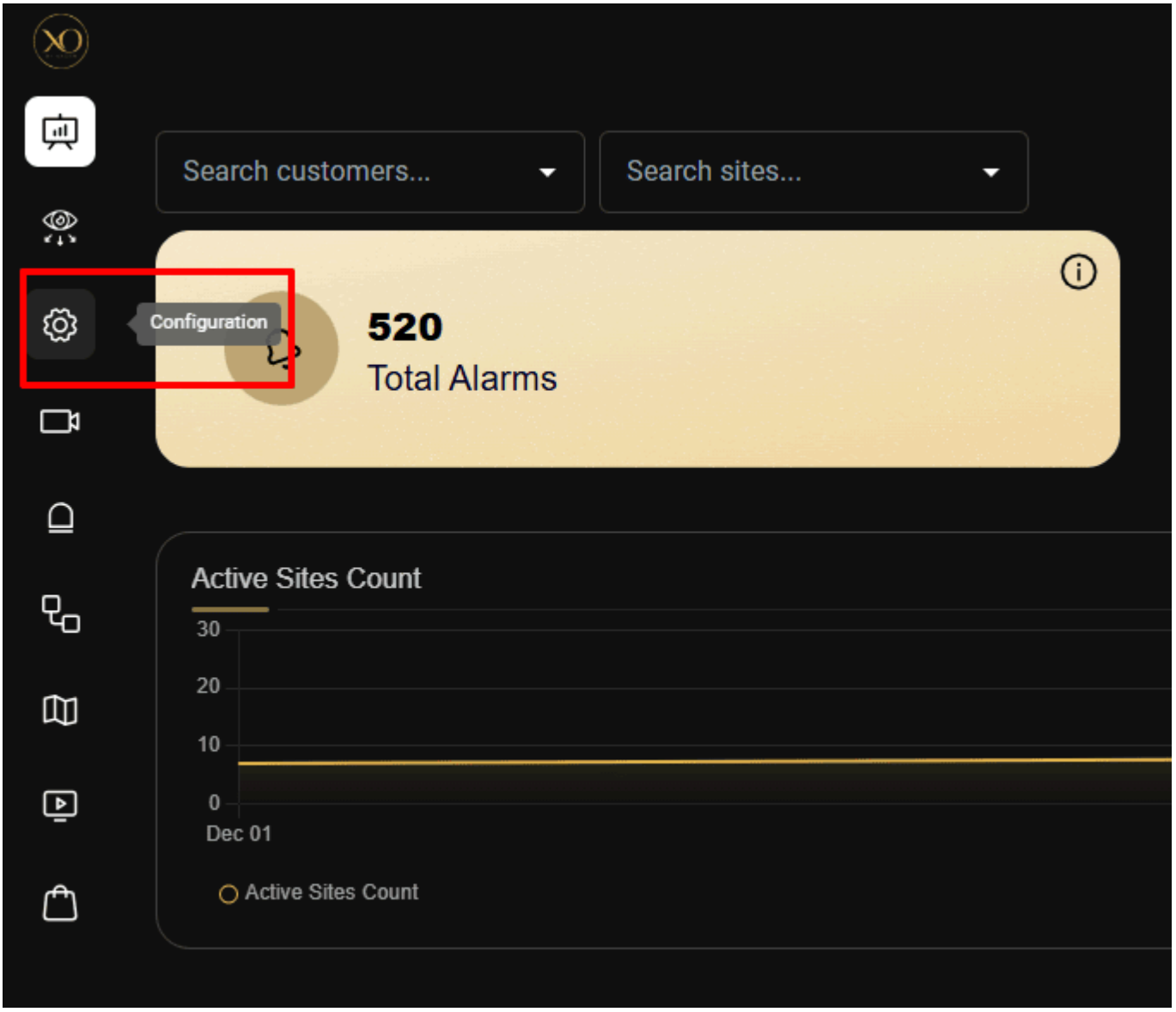


Step 3: Navigate to the Configuration App

From the left-hand **Apps** panel, locate the **Configuration** icon.

Click **Configuration** to open the configuration workspace.

The system will load the unified configuration screen, where all customer profiles, sites, and devices are managed.



Step 4: Understanding the Configuration Hierarchy

GCX-ONE follows a strict, intuitive hierarchy to ensure clear device ownership and maintain clean data structure: Customer → Site → Device → Cameras/Sensors.

How this hierarchy works

****Customer: ****Represents the top-level organizational entity (e.g., a company, enterprise, or client). Each customer contains one or more physical or logical sites.

****Site: ****Represents a physical location or deployment area (office, warehouse, tower site, retail store, etc.). Devices deployed at the location are associated with the site for clarity and operational control.

Axis IP Cameras

ACS Pro Recorders

Axis I/O Modules

Axis Horn Speakers

****Device: ****Represents the actual hardware endpoints such as:

This hierarchy ensures:

Clean configuration management

Accurate alarm routing

Multi-tenant separation

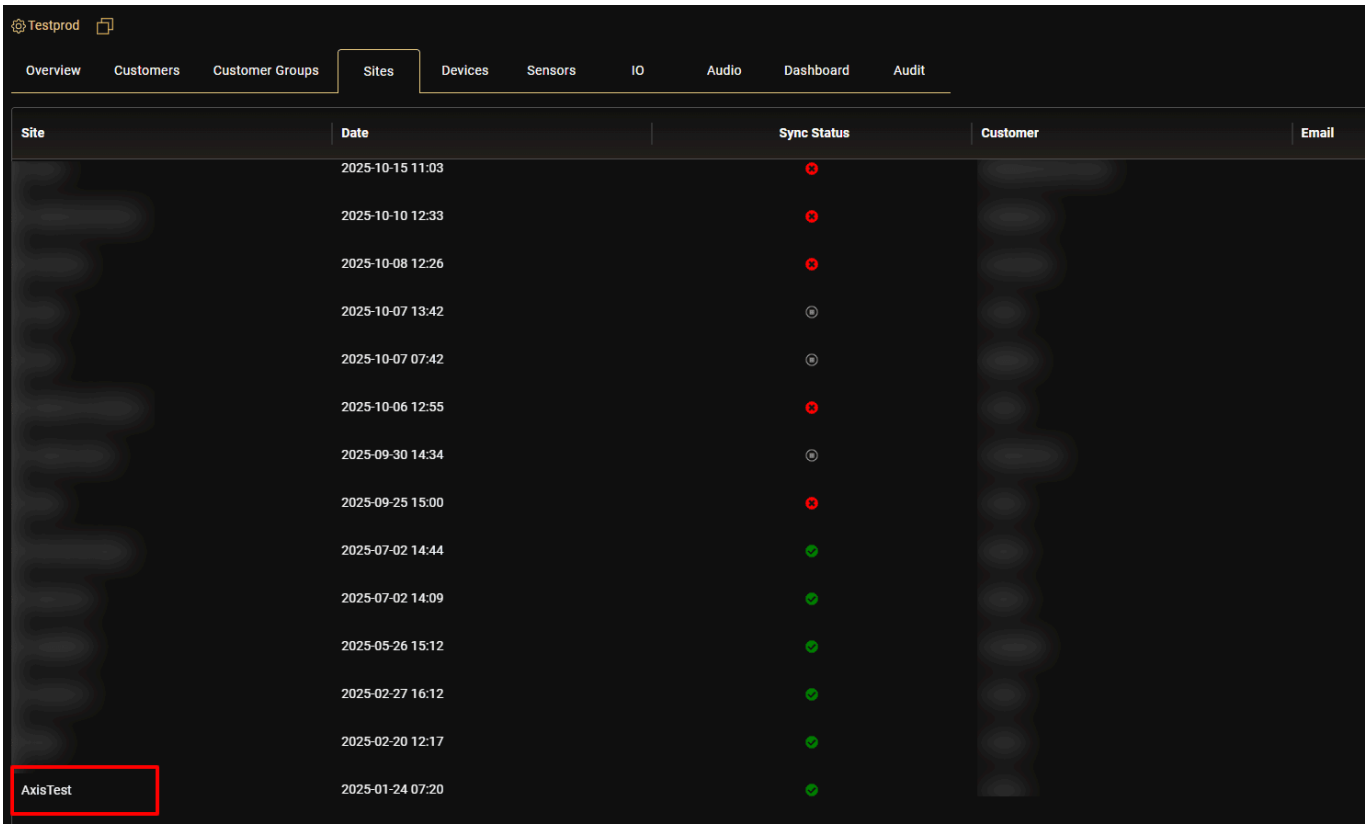
Site-wise reporting and visibility

Scalable onboarding for larger enterprise fleets

Step 5: Select or Create a Customer/Site

Once in the Configuration app:

Choose an existing customer/Site from the list, ****or ****Click ****Add New ****to create a new top-level customer and site profile.



Step 6: Add Devices to the Site

Once the site is created and selected:

Open the **Devices** tab under the site.

Click **Add New**.

Axis Camera Station (for ACS Pro devices)

Axis (for I/O Modules and IP Cameras)

GCX-ONE Audio (for Axis Horn Speakers)

Choose the type of Axis device you want to add:

The system will open the device configuration wizard.

Step 7: Provide Device details

GCX will prompt for:

Name

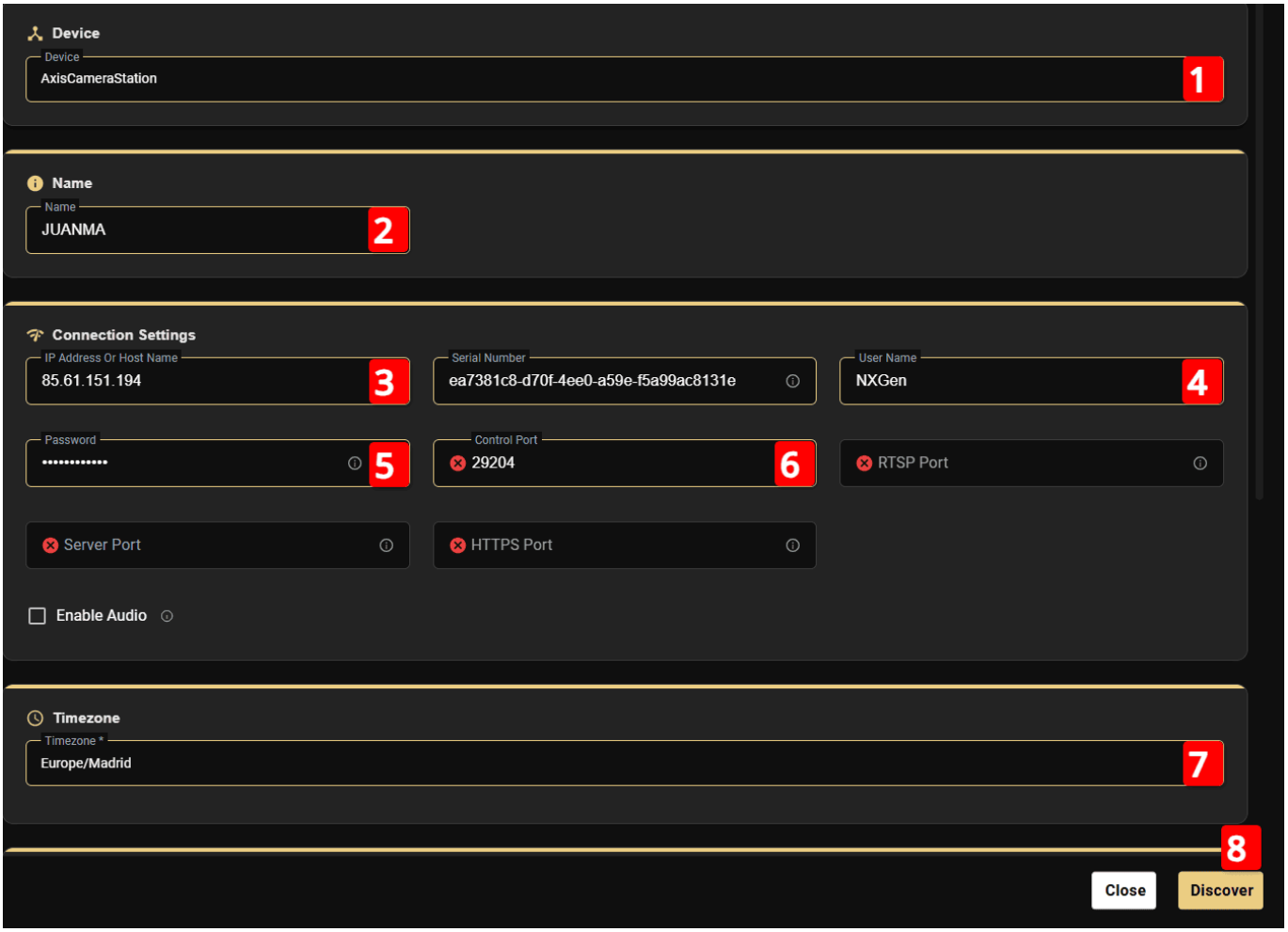
IP Address / Hostname

Control Port

Username & Password

Device Time zone

Once entered, click **Discover**.

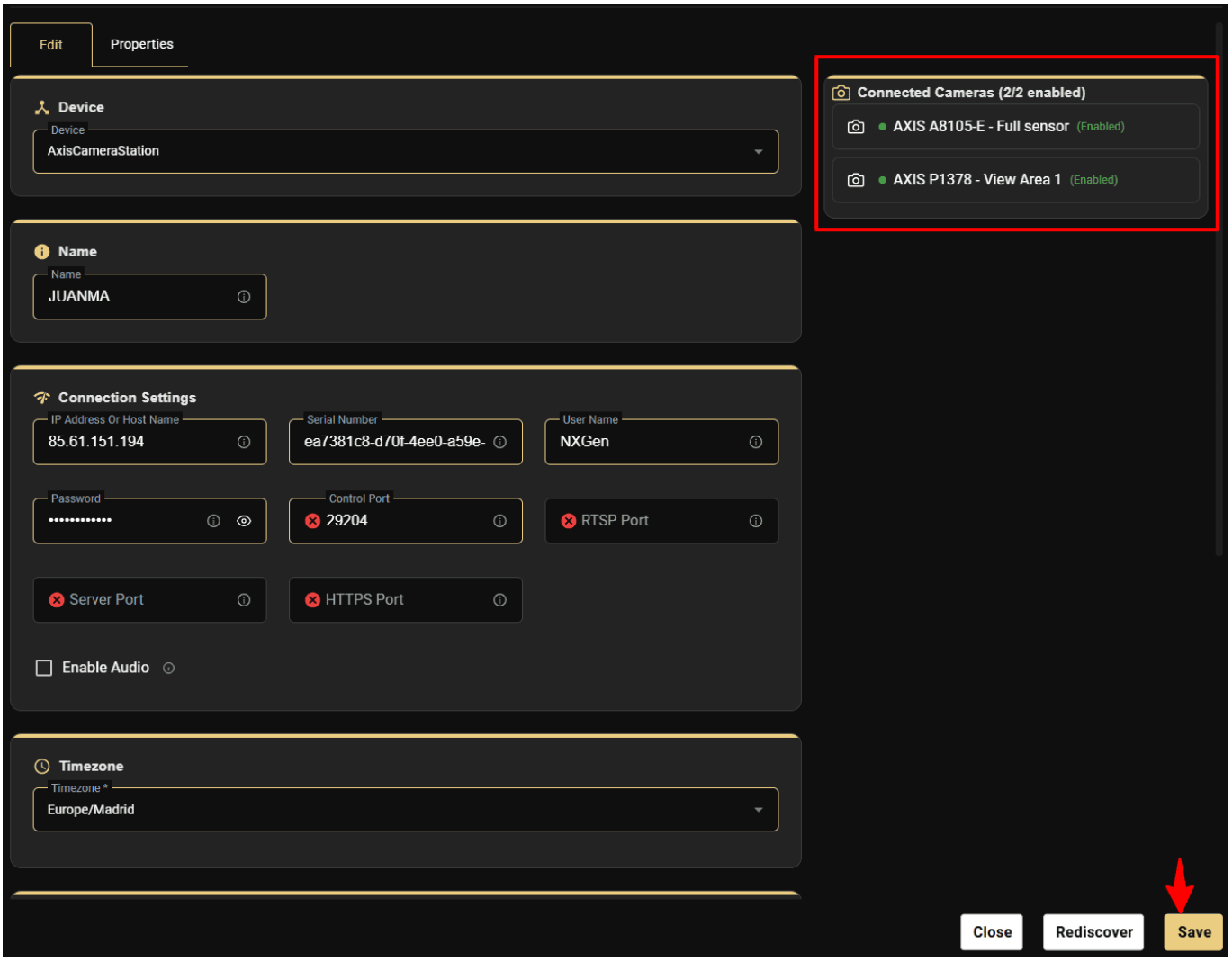


Step 8: Camera list discovery

GCX-ONE automatically retrieves:

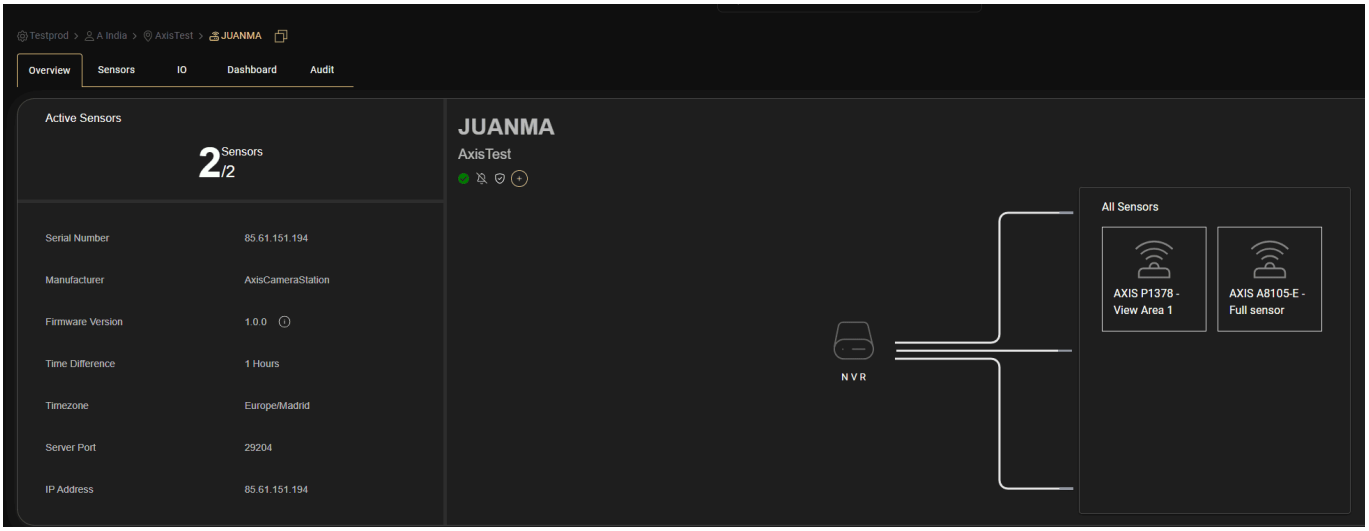
Camera list (for ACS Pro or IP Camera - if it has multiple profile/view)

I/O ports (for I/O modules)

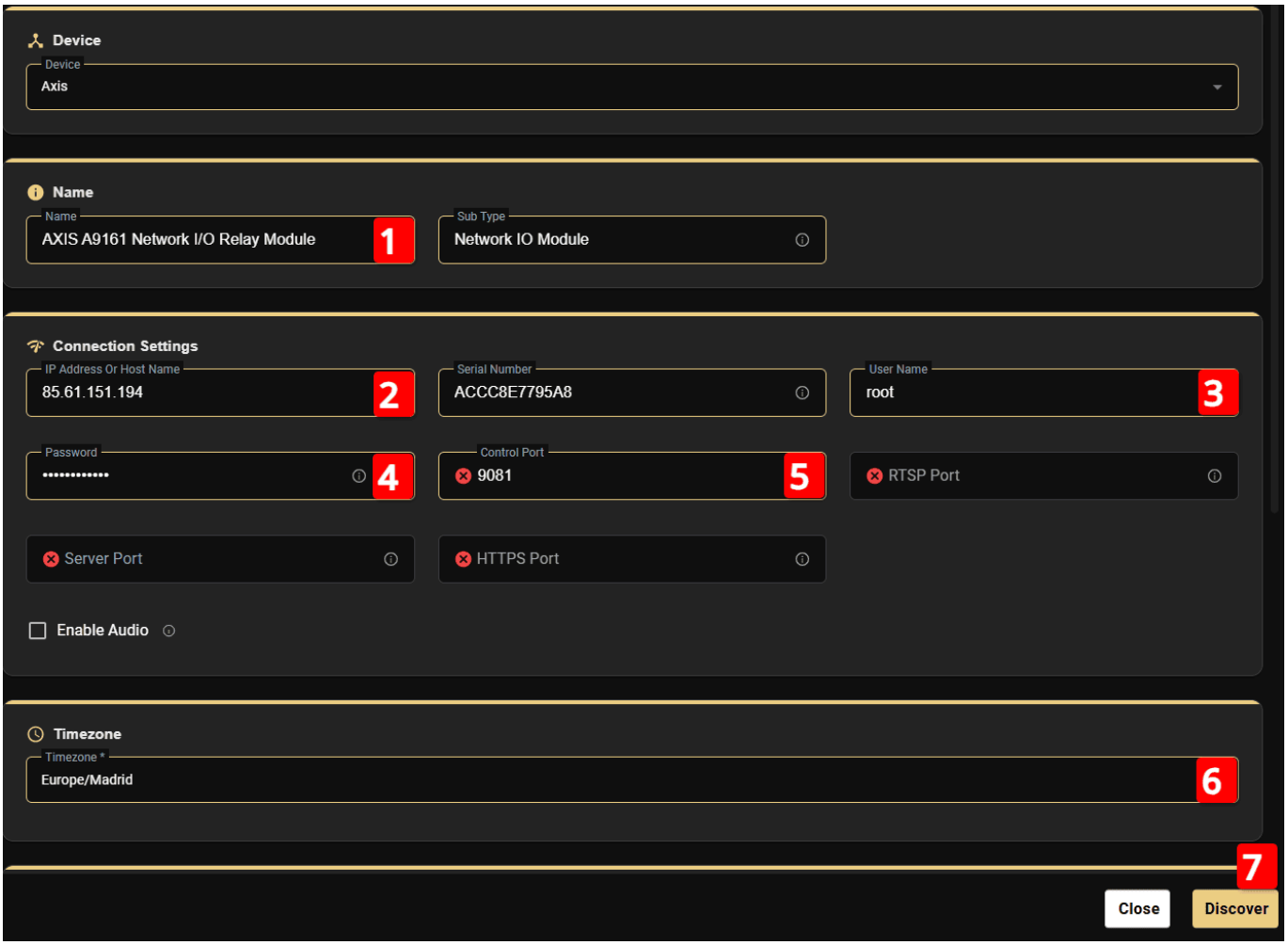


Click Save and complete the device addition.

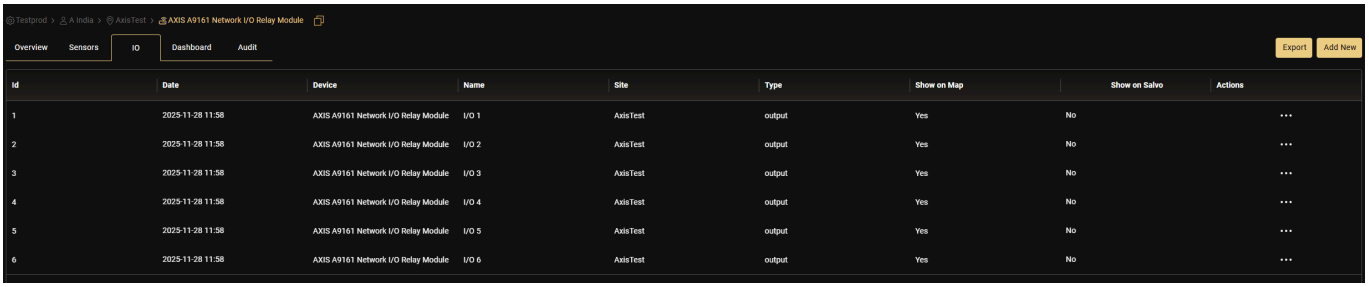
Device overview would be shown once the save is complete.



Similarly for Axis IO Modules, you can select Device Type as "Axis" and control port as the Vapix API port.

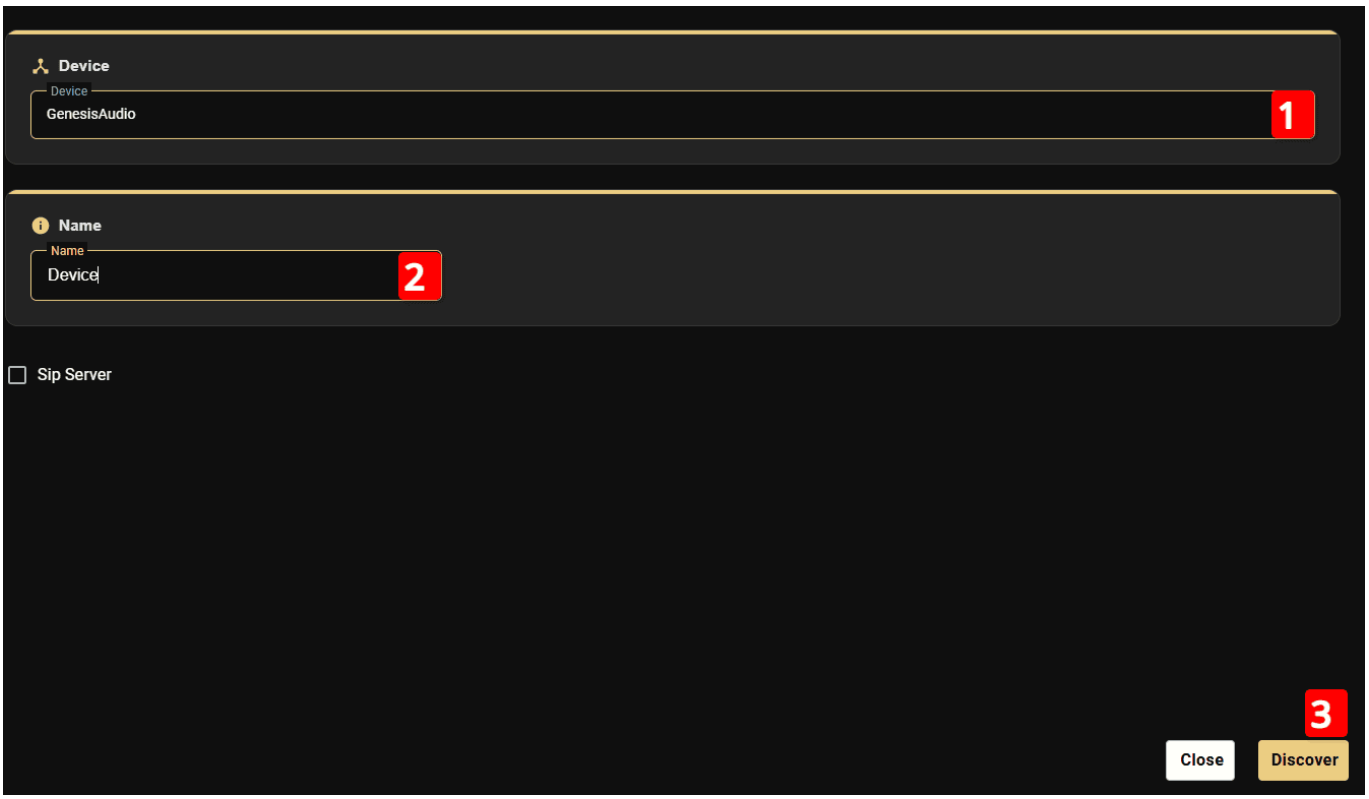


This will retrieve all available inputs and outputs and add them to the device I/O list.



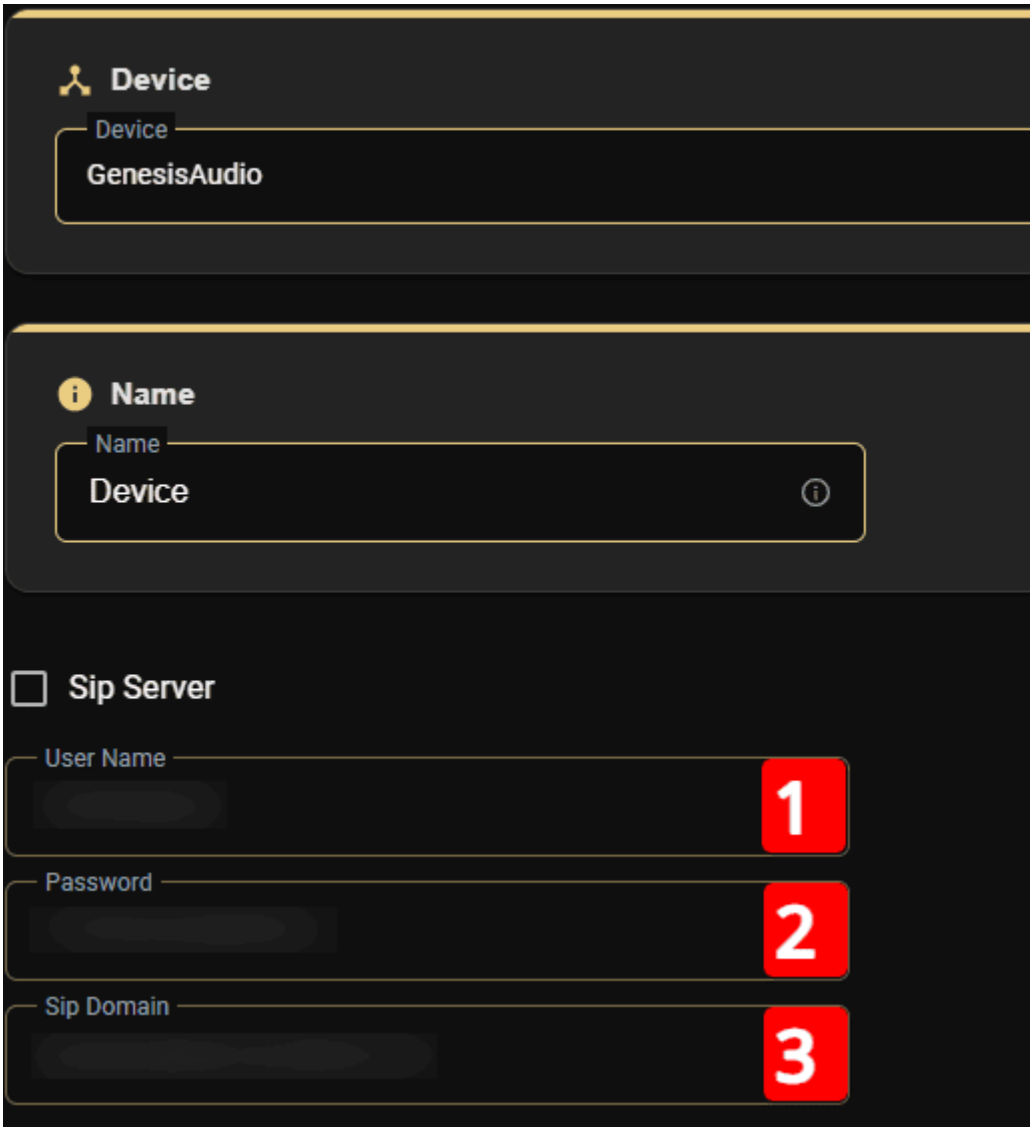
Id	Date	Device	Name	Site	Type	Show on Map	Show on Salvo	Actions
1	2025-11-28 11:58	AXIS A9161 Network I/O Relay Module	I/O 1	AxisTest	output	Yes	No	•••
2	2025-11-28 11:58	AXIS A9161 Network I/O Relay Module	I/O 2	AxisTest	output	Yes	No	•••
3	2025-11-28 11:58	AXIS A9161 Network I/O Relay Module	I/O 3	AxisTest	output	Yes	No	•••
4	2025-11-28 11:58	AXIS A9161 Network I/O Relay Module	I/O 4	AxisTest	output	Yes	No	•••
5	2025-11-28 11:58	AXIS A9161 Network I/O Relay Module	I/O 5	AxisTest	output	Yes	No	•••
6	2025-11-28 11:58	AXIS A9161 Network I/O Relay Module	I/O 6	AxisTest	output	Yes	No	•••

For Axis Horn Speaker, you can select the Device type as "GCX-ONE Audio" and give a valid name and hit discover.



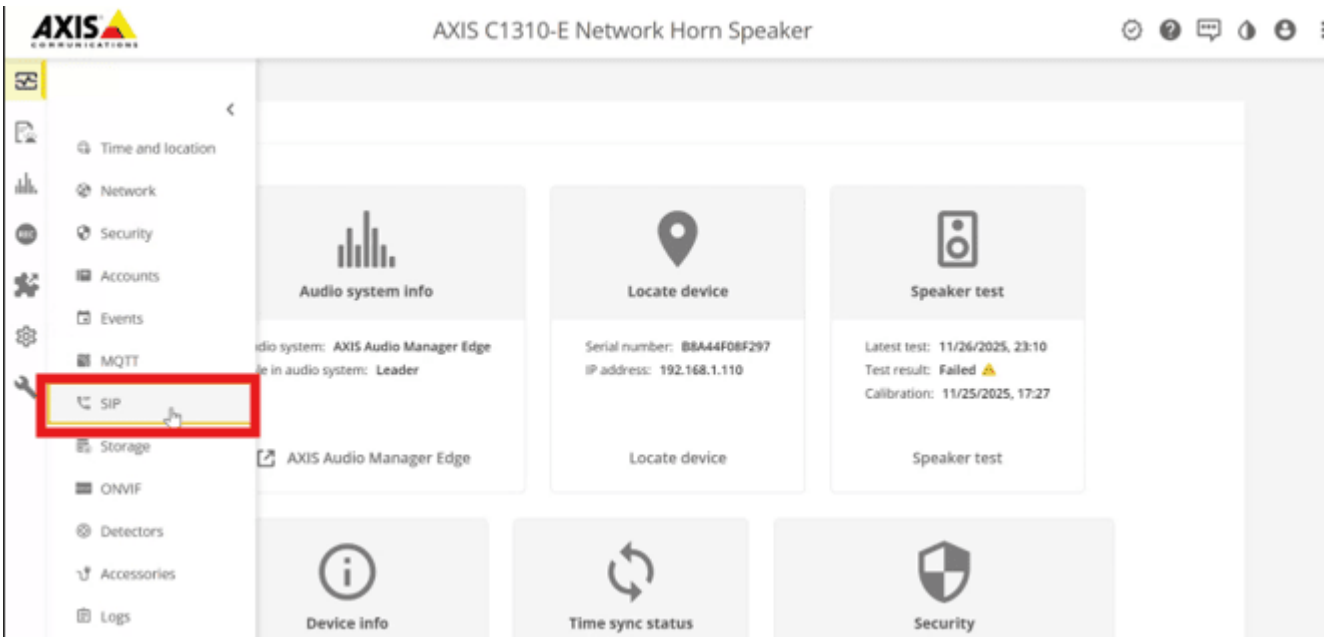
After the horn speaker is discovered and added, GCX-ONE automatically generates the SIP username, password, and domain details. These credentials must be configured on the horn speaker so it can register with the SIP endpoint and enable audio

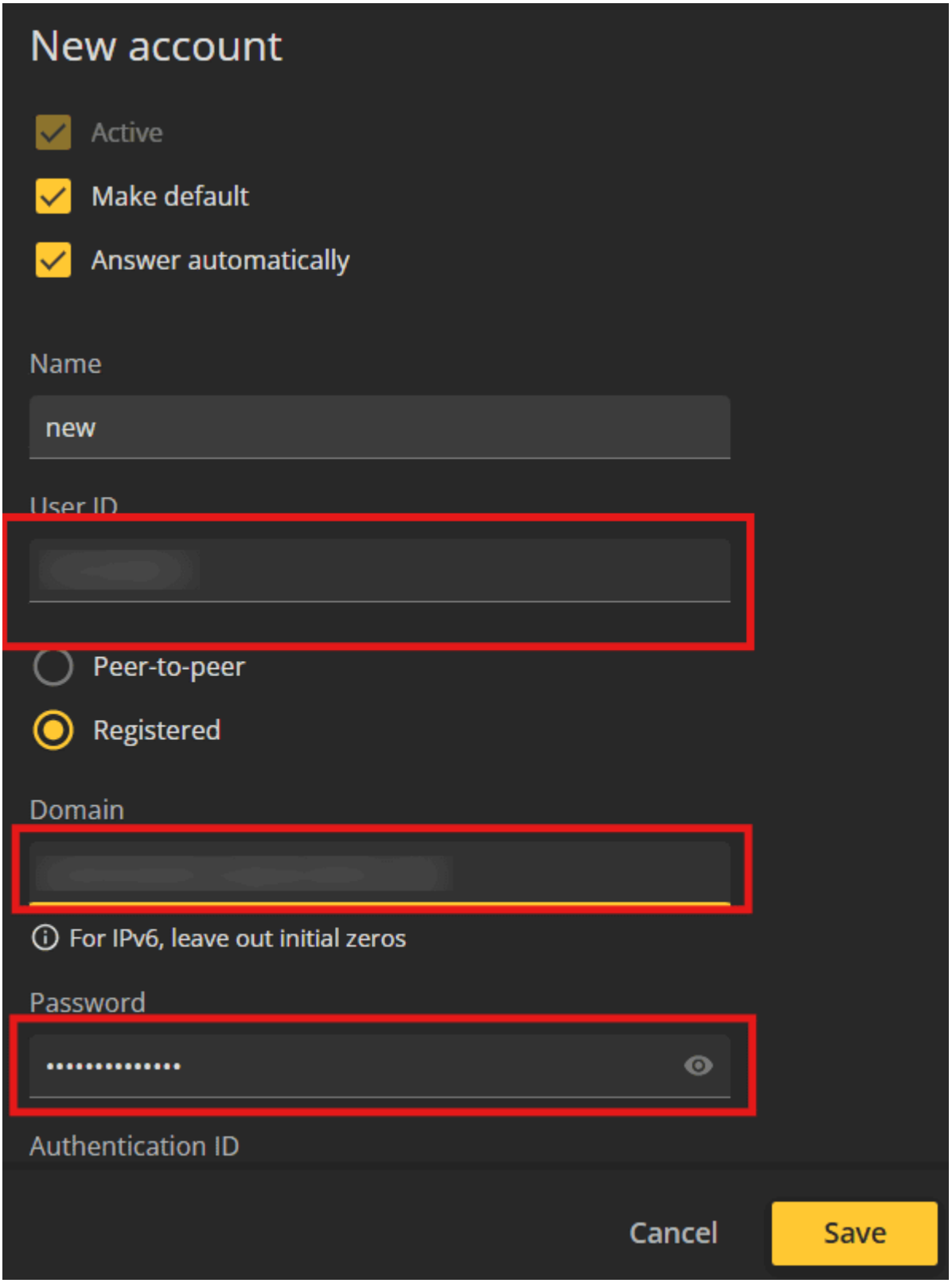
communication.



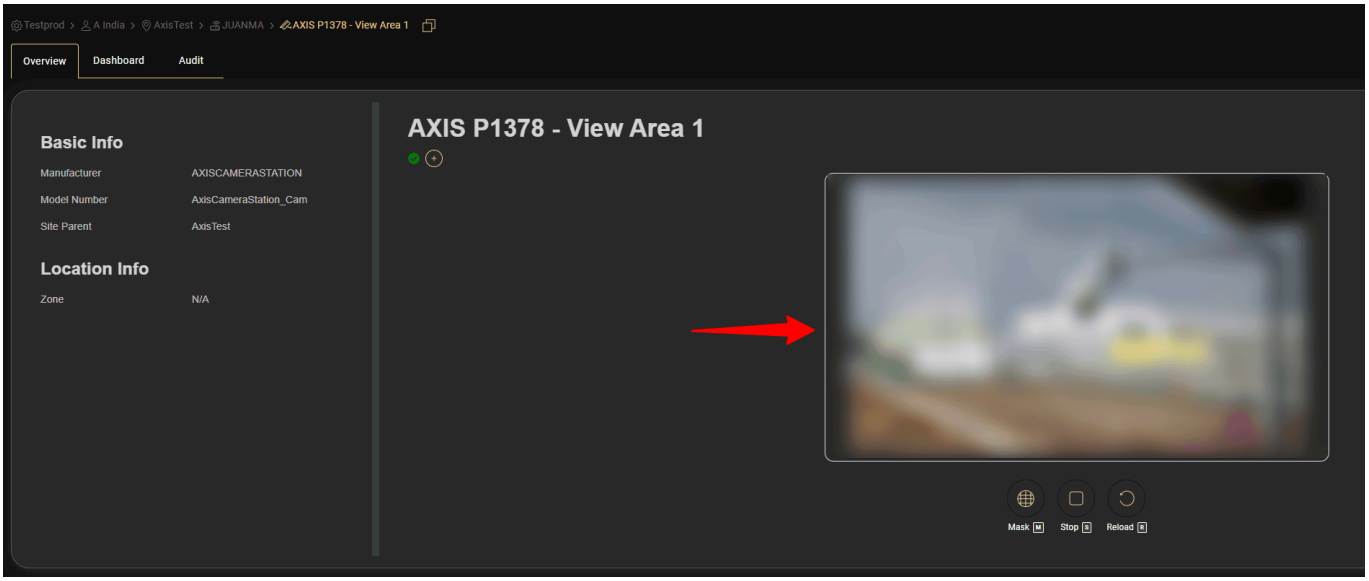
Make sure the above SIP details are configured in the horn speaker by entering the generated username, password, and domain into the device's SIP settings page and saving the configuration. This ensures the speaker successfully registers with

the GCX-ONE SIP endpoint for audio communication.

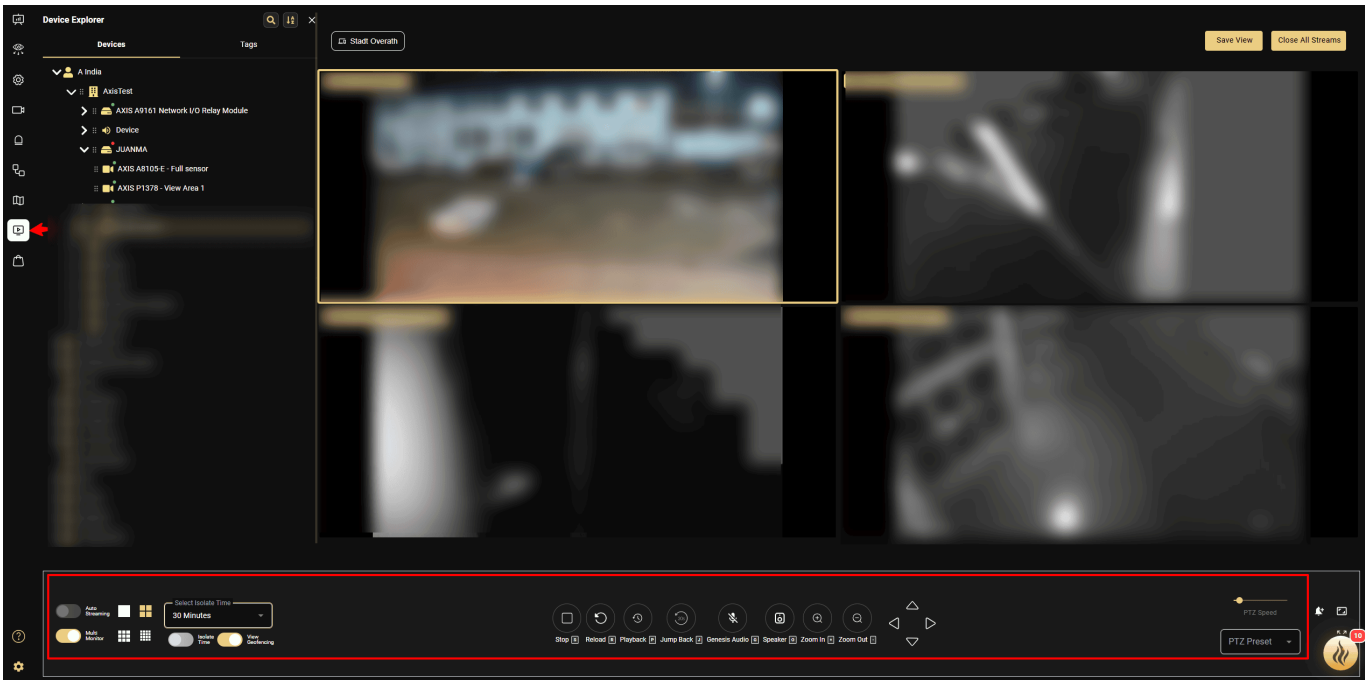




To verify that the camera is streaming correctly, open any camera from the configured device list. You should be able to view the live video stream without interruptions.



In the Video Viewer application, you can perform additional video operations such as multi-camera streaming, PTZ control, preset management, playback of recorded footage, and more. Simply drag and drop the device into the play area, and the system will begin displaying the live multi-camera stream.



Events and Rules Configuration

Login to the **ACS Client**.

Navigate to the **Events** section and open **Action Rules**.

Click **New Action Rule** to start the configuration wizard.

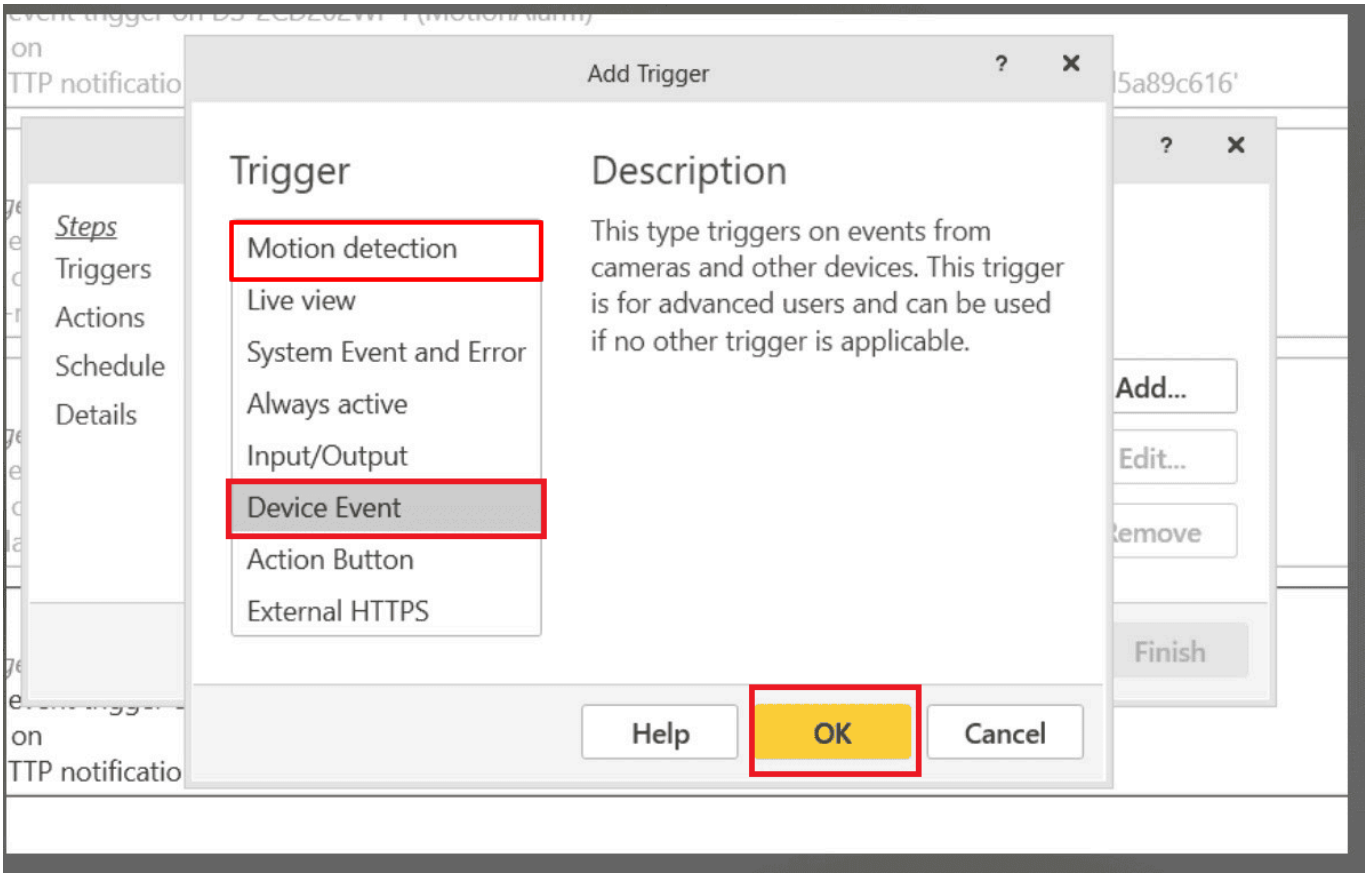
Motion detection

Device events

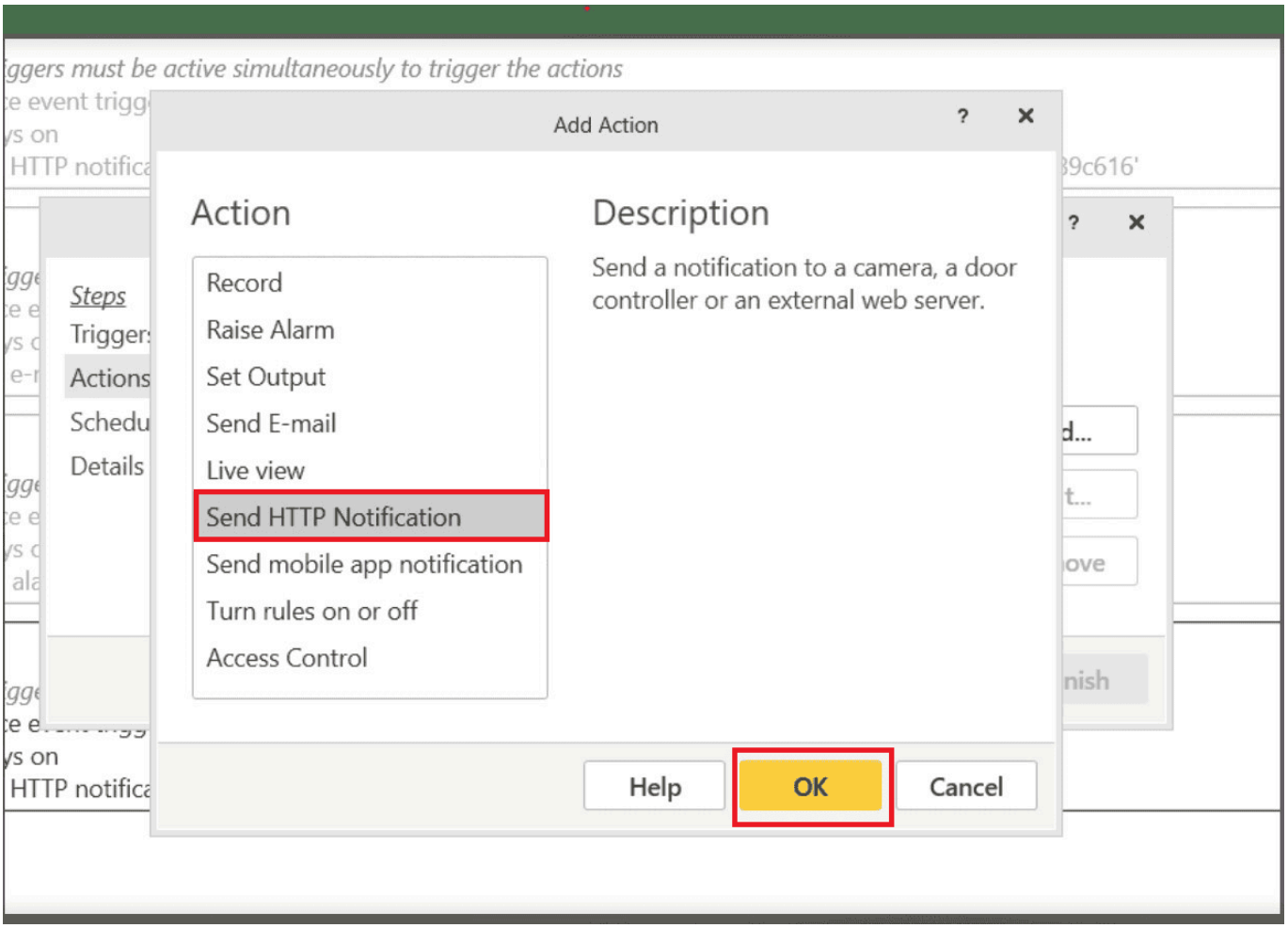
Input/Output events

Choose the desired **alarm trigger**, such as:

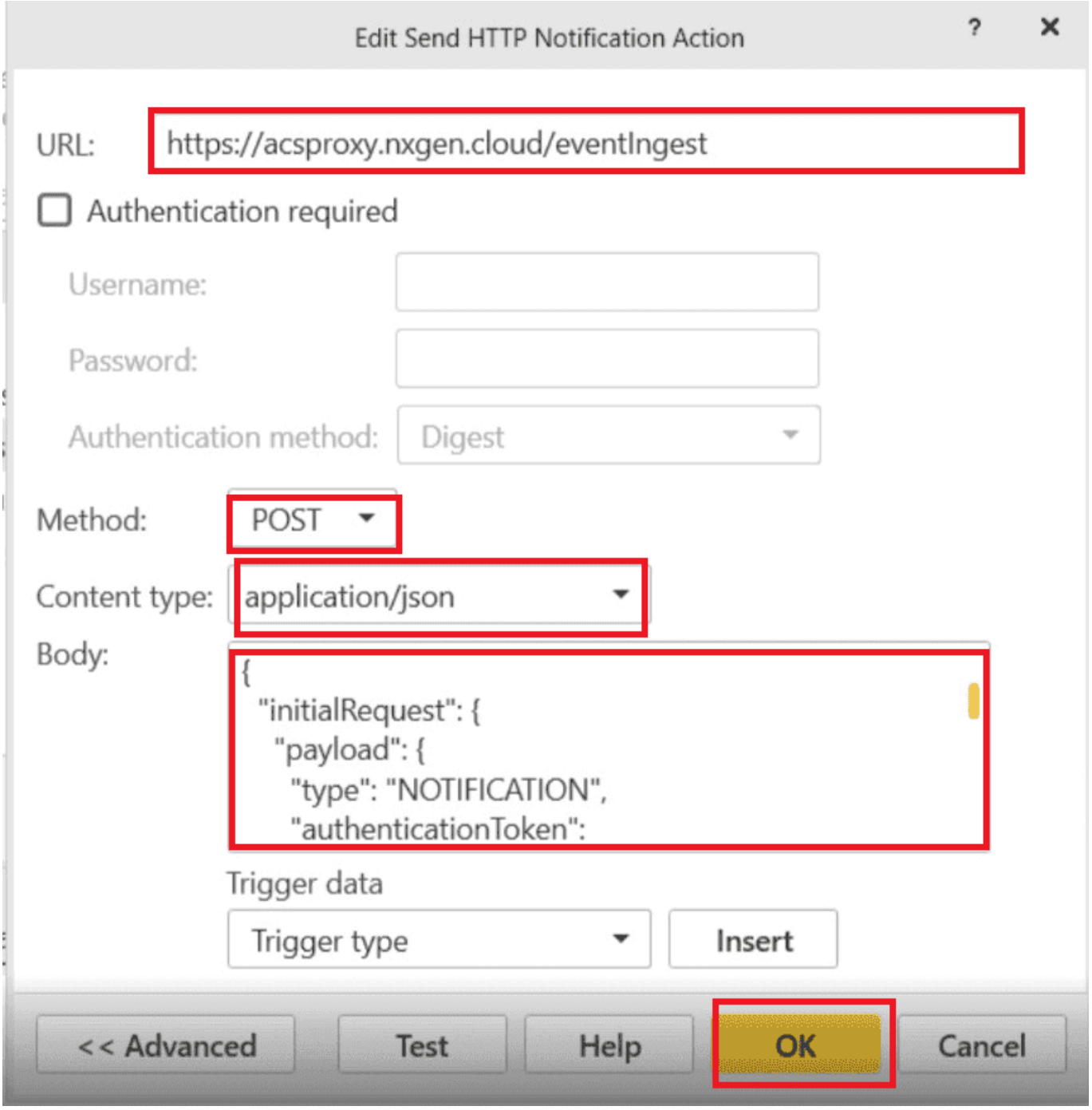
GCX-ONE supports alarm forwarding from ACS pro through **Action Rules**. To configure this:



After selecting the trigger, set the **Action** to **“Send HTTP Notification”**.



Proceed with the wizard to define notification details.



Configure the following parameters in the HTTP Notification:

{

"initialRequest": {

```
"payload": {

  "type": "NOTIFICATION",

  "authenticationToken": "<<DEVICE_ID_FROM_GENESIS>>",
```

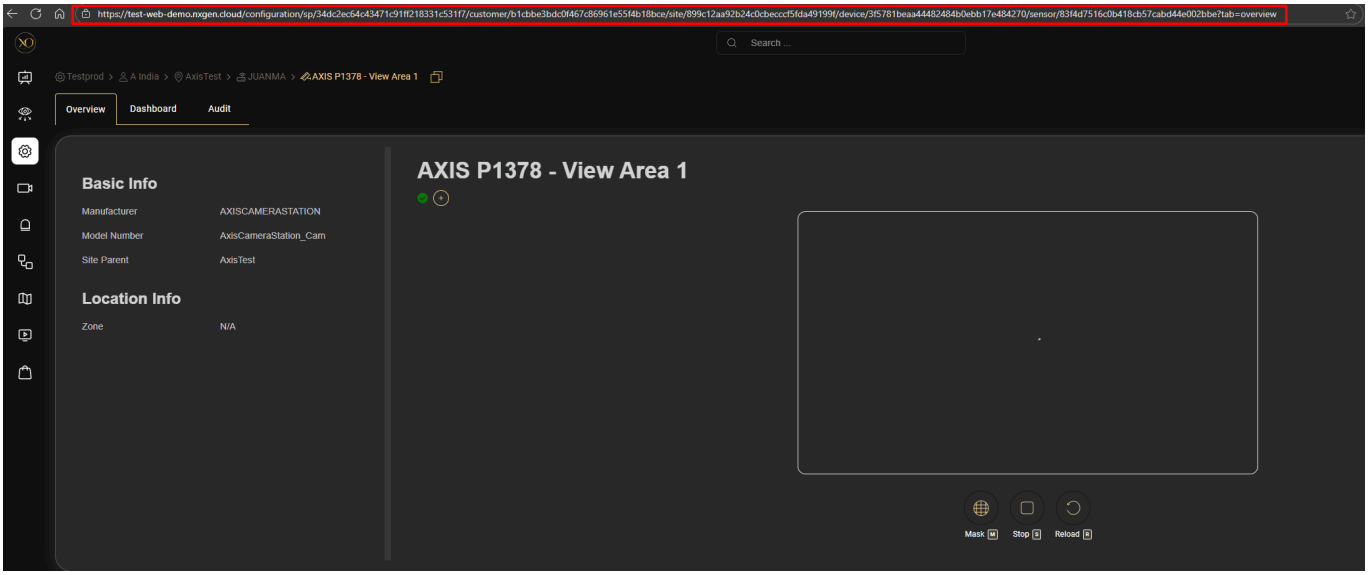
```
  "notifications": [
    {
      "event": {
        "type": "ALARM_TRIGGERED",
        "genesisCamId": "<<CAMERA_ID_FROM_GENESIS>>",
        "cameraId": "$(TriggerData.SourceId)",
        "timestamp": "$(TriggerData.TimeUtc)"
      }
    }
  ]
}
```

}

}

To configure the Request Body, Use the following JSON template. Replace the placeholder values (<>, <>) with the actual device id and camera id which you can get it from the GCX-ONE Configuration URL.

You can obtain the placeholder values directly from the configuration URL.



Here is how you can interpret the URL to extract the placeholder values for the Device ID and Camera ID.

Testing and Verification

Trigger a known rule in ACS Pro (e.g., motion or analytics alert).

In GCX-ONE, verify that the corresponding event appears in Video activity search or device level alarm dashboard.

If no events appear, confirm network reachability to https://acsproxy.nxgen.cloud/eventIngest and verify that the Action Set is active.

Best Practices

Always test connectivity with https://acsproxy.nxgen.cloud/eventIngest before saving.

Ensure firewall or outbound network policies allow TCP traffic to the GCX-ONE endpoint.

Avoid modifying placeholder syntax — any incorrect character will break JSON parsing.

Alarm Processing Overview

GCX-ONE provides a multi-application alarm handling workflow designed for professional monitoring centers. When an alarm is received from an Axis device, three key applications work together to give operators complete situational awareness and rapid response capability.

1. Workflow Application (Alarm Handling)

The Workflow app (typically the Talos alarm management view) acts as the operator's primary alarm console.

Alarms are listed with full details including device, site, customer, timestamps, trigger type, and severity.

Operators open an alarm to begin investigation, acknowledge it, add notes, assign it, or escalate it as required.

This application serves as the central control point for managing the entire alarm lifecycle.

2. Video Viewer Application (Alarm Verification)

Pre-alarm image

Current-alarm image

Post-alarm image.

When an alarm is opened in the Workflow app, the Video Viewer automatically switches to the corresponding alarm context. It displays:

Live Video

Playback Timeline

PTZ Control

Preset Navigation

Audio Announcement/Talk-down

Multi-camera layouts

Operators also have access to a full suite of video tools:

This gives operators immediate visual confirmation and the ability to verify the event in real time.

3. Map View (Geospatial Awareness & Output Control)

The Map application displays all devices and outputs positioned on an interactive site map.

When an alarm is selected in the Workflow app, the Map view automatically centers on the alarm location.

Sirens

Strobes

I/O Relays

Recorder audio clip trigger through Horn Speaker.

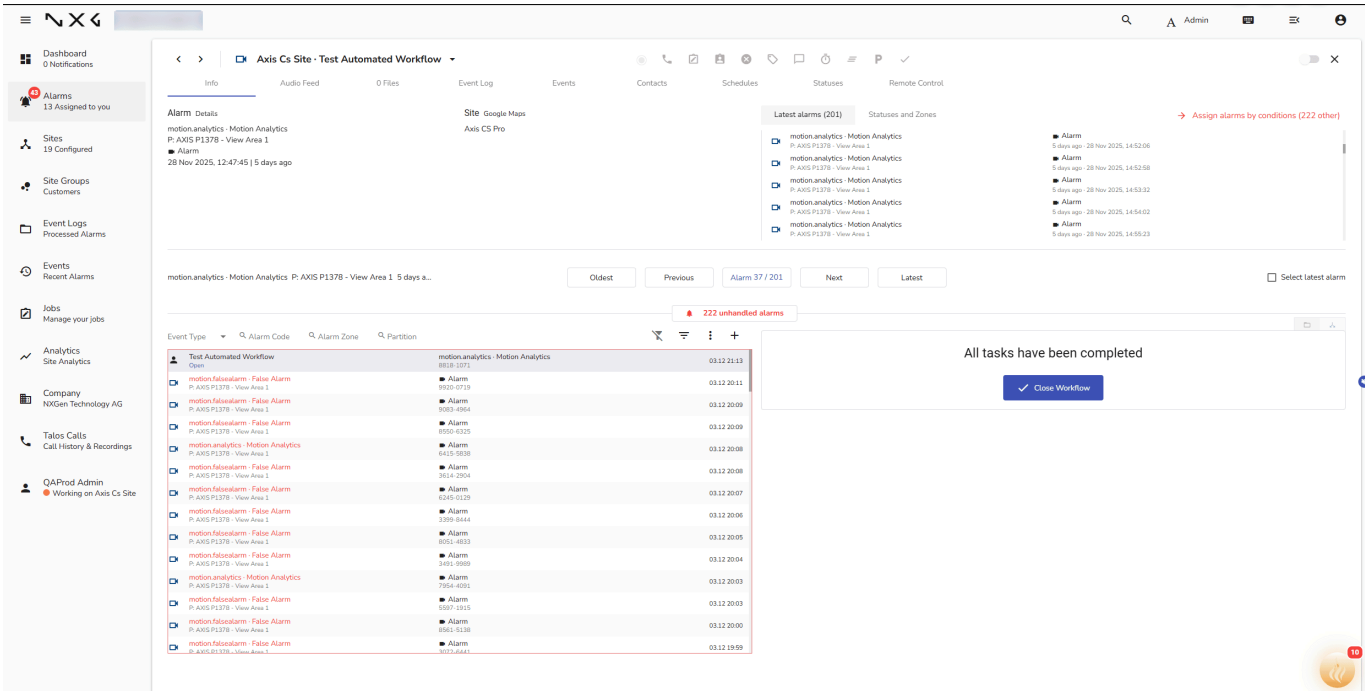
Operators can interact with devices on the map and trigger outputs such as:

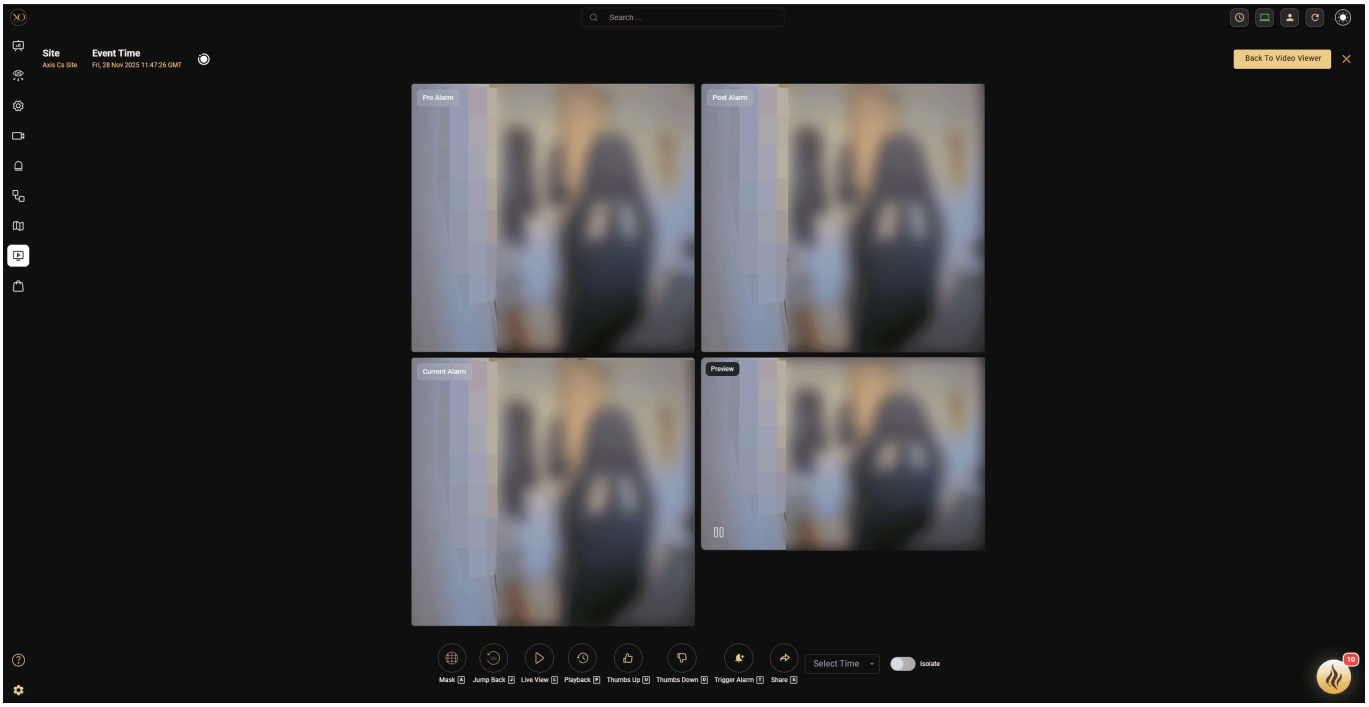
This gives operators fast access to response actions directly on the site layout.

Multi-Monitor Operator Workflow

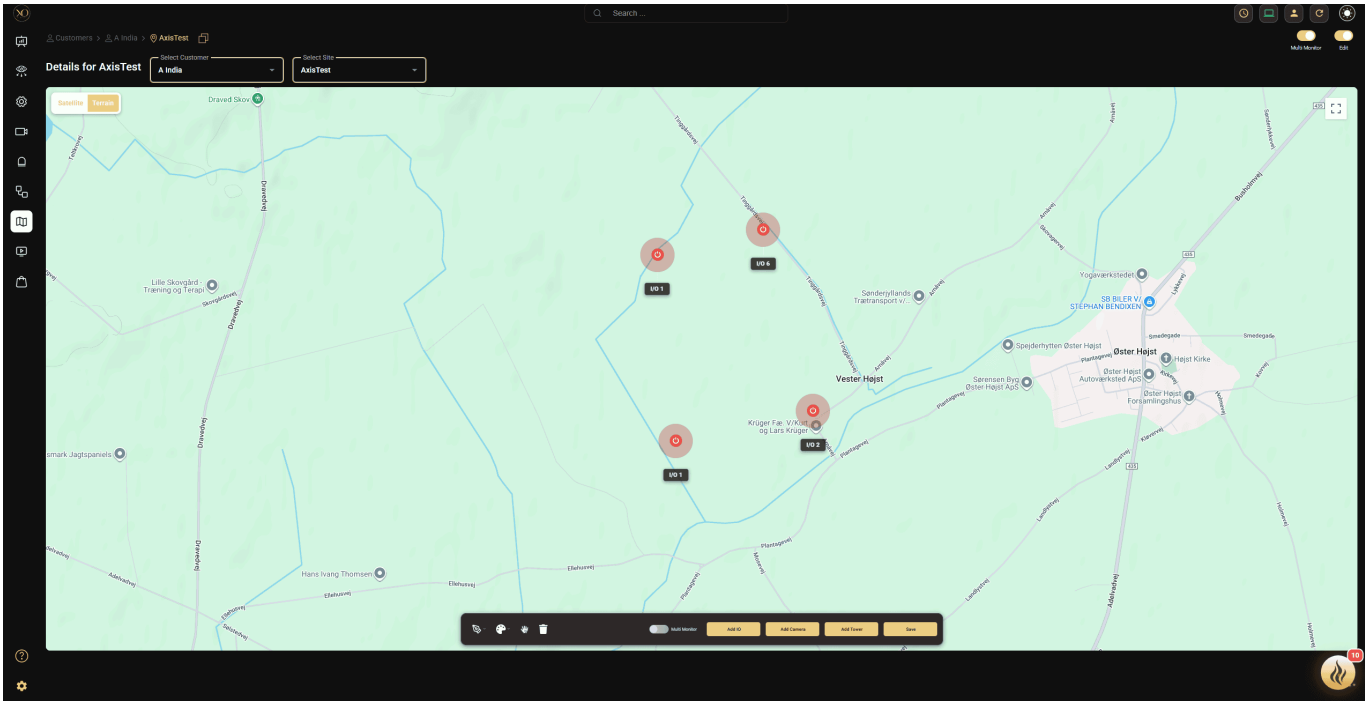
In professional control room setups, operators typically use three or more monitors. GCX-ONE is designed to support this environment:

Monitor 1: Workflow application





Monitor 2: Video Viewer



Monitor 3: Map View

When the operator opens an alarm in the Workflow application:

The Video Viewer on its dedicated monitor automatically switches to the alarm's video source(s).

The Map View simultaneously updates to highlight the alarm's site and device location.

This synchronized context switching ensures that:

The operator has a holistic, 360° view of the alarm.

All relevant tools (video, PTZ, audio, maps, outputs) are instantly accessible.

Response time is minimized, and situational understanding is maximized.

Integrations Features Overview

The GCX-ONE platform integrates with Axis Camera Station Pro through a combination of RESTful APIs, RTSP streaming, and webhook-based event forwarding. This integration enables unified management, monitoring, and analytics for all connected cameras.

Summary

By following this guide, you can integrate **Axis Camera Station Pro** and the full Axis ecosystem into your **NXGEN monitoring platform**.This integration unifies:

Live and recorded video

Alarm ingestion

Audio announcements

I/O control

Multi-application operator workflows

Resulting in **comprehensive, scalable, and efficient monitoring operations.**